

DHCP - Dynamic Host Configuration Protocol

*Computer Networking: A
Top-Down Approach*

9th edition

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1 What Is DHCP?

• Definition

DHCP is a network management protocol that automatically assigns IP configuration to devices on a network.

Instead of manually configuring:

- IP address
- Subnet mask
- Default gateway
- DNS server

DHCP does it automatically.

2 Why DHCP Is Needed

Without DHCP:

- Manual IP configuration
- Error-prone
- Duplicate IPs
- Hard to manage large networks

With DHCP:

- ✓ Plug-and-play
- ✓ No conflicts
- ✓ Centralized management

3 What Information DHCP Provides

A DHCP server typically assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time

4 Where DHCP Runs

- DHCP Client: End device (PC, phone)
 - DHCP Server: Router or dedicated server
- 📌 Home routers usually act as DHCP servers.

5 DHCP Working Principle (DORA Process)

DHCP works in four steps, called DORA.

Step-by-Step DHCP Process (DORA)

1 DHCP Discover

 Client broadcasts:

“Is there any DHCP server?”

nginx

 Copy code

Src IP: 0.0.0.0

Dst IP: 255.255.255.255

Why broadcast?

- Client has no IP yet

2 DHCP Offer

 Server replies with an offer:

- Available IP
- Subnet mask
- Gateway
- DNS
- Lease time

The offer is usually broadcast, since client has no IP yet.

3 DHCP Request

 Client responds:

“I accept this offer”

- Confirms selected server
- Declines other offers

4 DHCP Acknowledge (ACK)

✓ Server finalizes assignment:

- IP officially leased
- Client can now communicate normally

6 DHCP Lease Concept

- IP is leased, not permanent
- Lease has a time limit

Renewal:

- Client renews lease at 50% time
- If server unavailable, tries again later

7 DHCP Example (Real Life)

You connect your phone to Wi-Fi:

1. Phone broadcasts DHCP Discover
2. Router offers 192.168.1.20
3. Phone requests it
4. Router acknowledges
 - ✓ Internet works

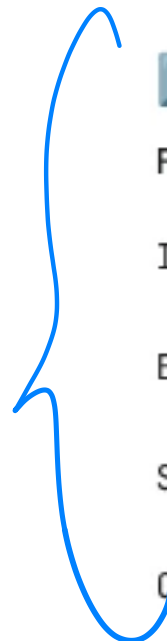
8 DHCP Ports (Exam Favorite ★)

Direction	Protocol	Port
Client → Server	UDP	67
Server → Client	UDP	68



9 DHCP vs Static IP

Feature	DHCP	Static
IP assignment	Automatic	Manual
Errors	Rare	Common
Scalability	High	Low
Control	Centralized	Per device



DHCP: Dynamic Host Configuration Protocol

goal: host obtains IP address from network server when it "joins" network

- [redacted]
- [redacted]
- [redacted]

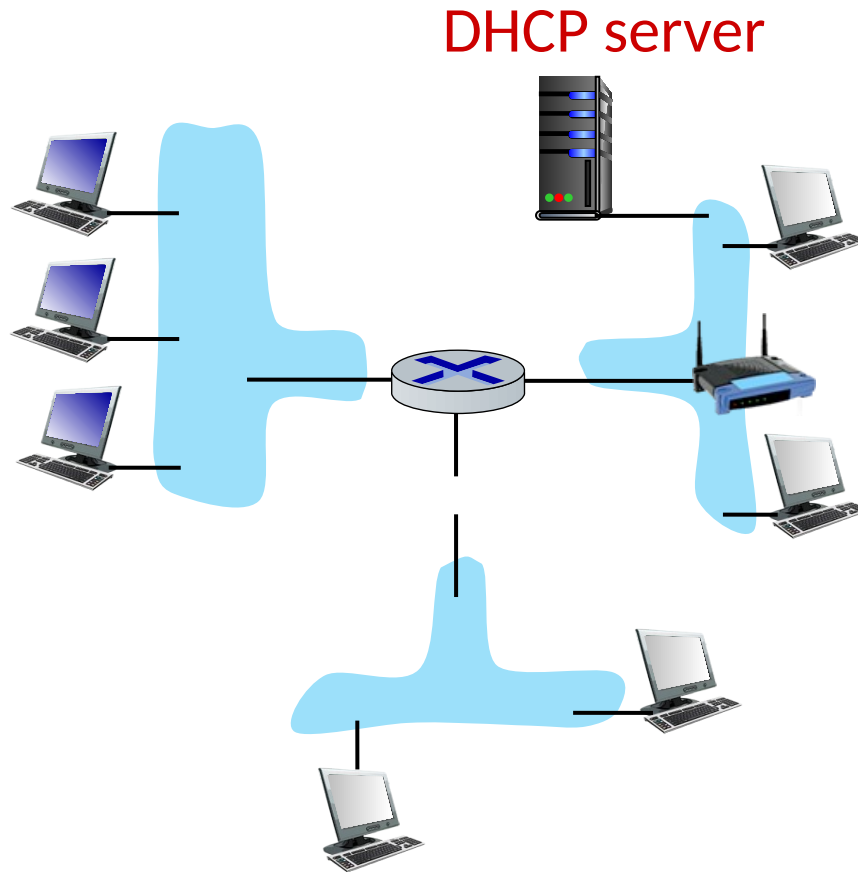


DHCP overview:

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg

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DHCP client-server scenario



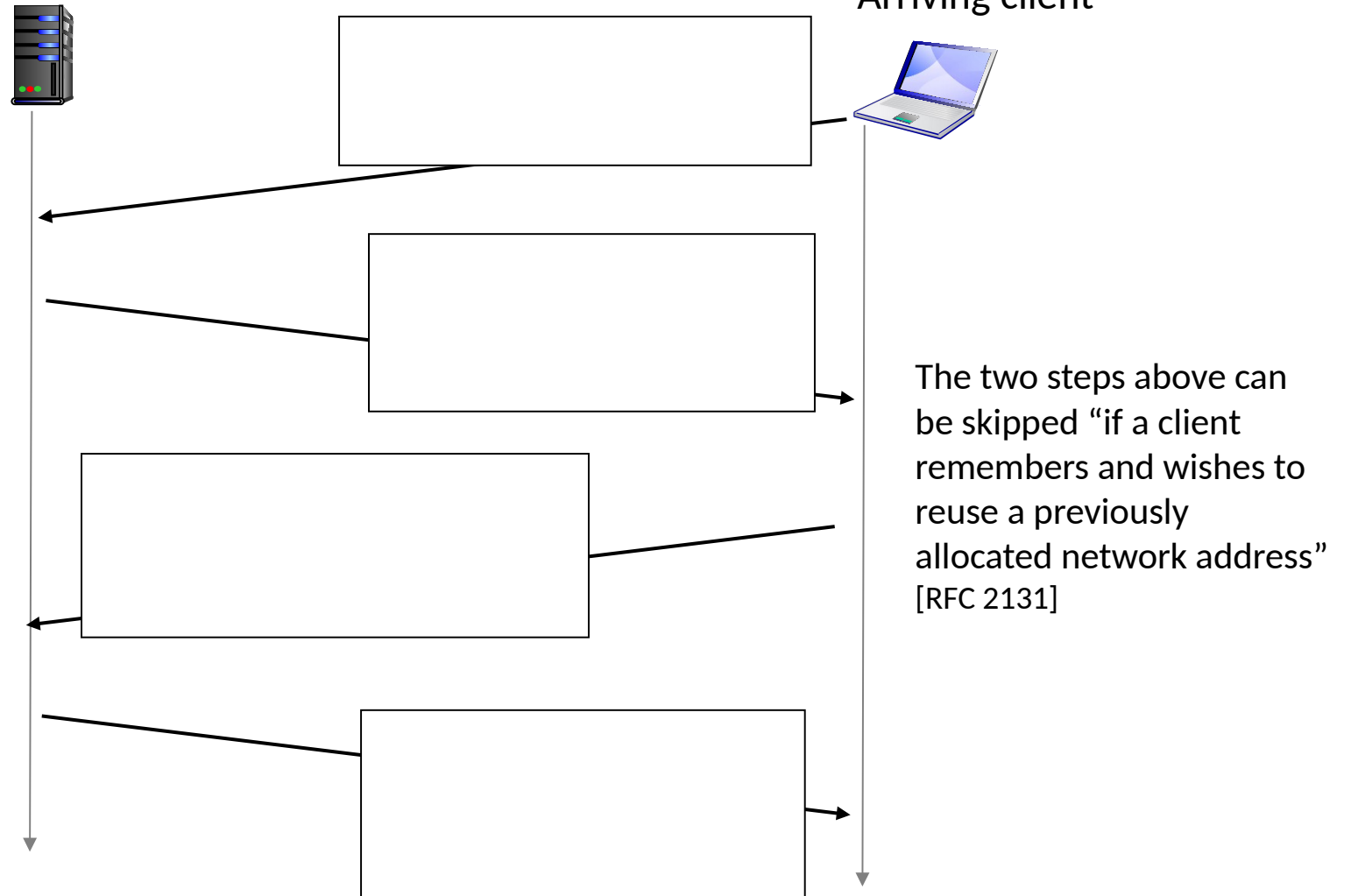
Typically, DHCP server will be co-located in router, serving all subnets to which router is attached

arriving **DHCP client** needs address in this network

DHCP client-server scenario

DHCP server: 223.1.2.5

Arriving client

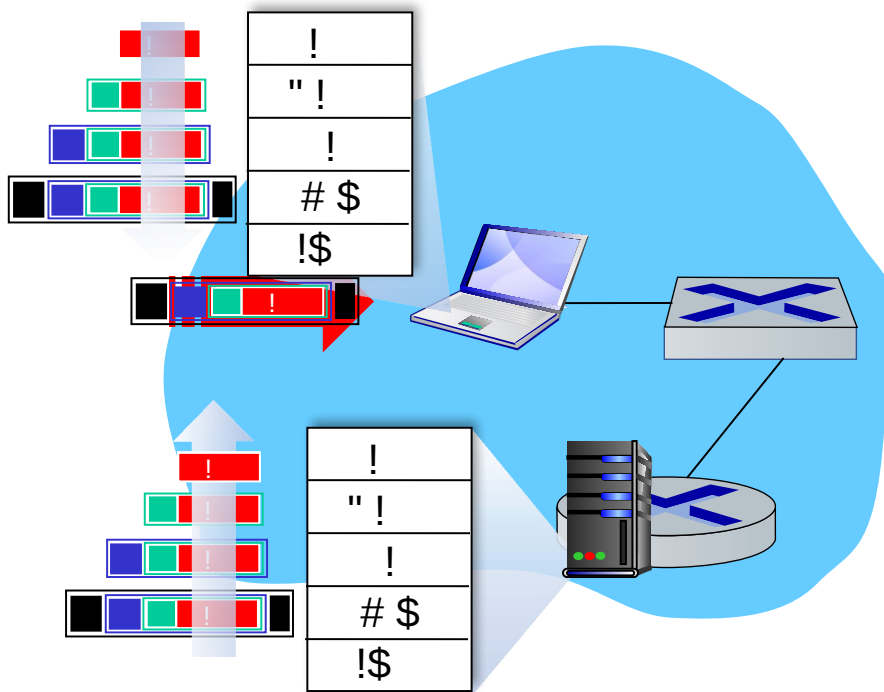


DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

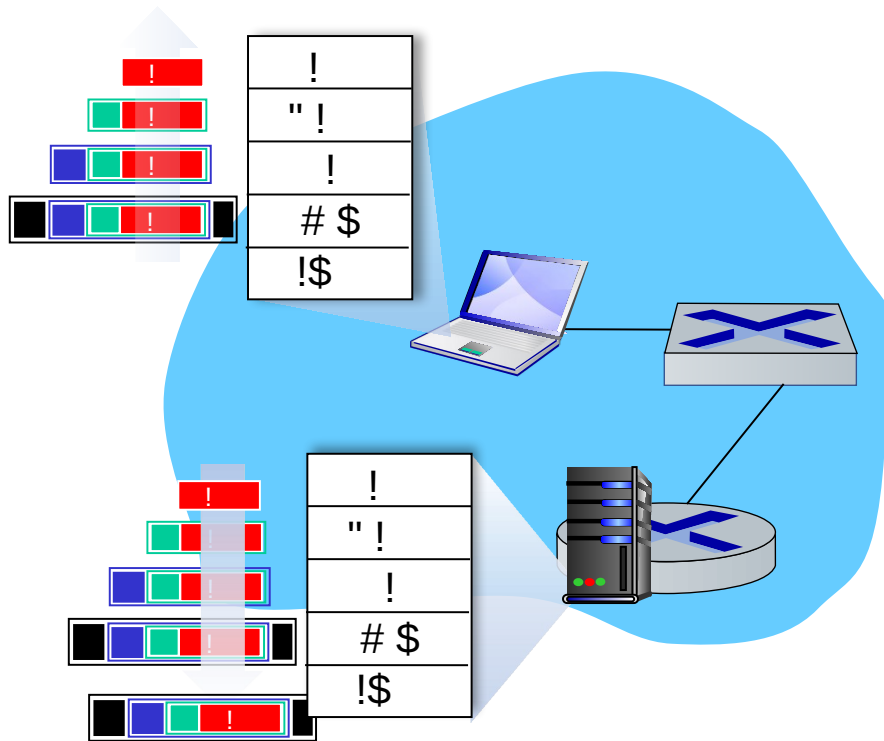
- ✓ address of first-hop router for client
- ✓ name and IP address of DNS sever
- ✓ network mask (indicating network versus host portion of address)

DHCP: example



- ✓ Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- ✓ DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- ✓ Ethernet frame broadcast (dest: FFFFFFFF) on LAN, received at router running DHCP server
- ✓ Ethernet de-mux'ed to IP de-mux'ed, UDP de-mux'ed to DHCP

DHCP: example



- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, de-muxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

DHCP: Which layer protocol?

- Not a Network layer protocol
- Why not?
- Is a Application layer protocol.
- Why it is a application layer protocol?
 - Defines message formats

? Why is DHCP an Application Layer protocol?

Even though DHCP:

- Uses UDP
- Deals with IP addresses
- Interacts closely with the network layer

...it is still classified as an Application Layer protocol.

🔑 Short Answer (Exam-Perfect)

DHCP is an application-layer protocol because it provides a network configuration service to applications and hosts, runs in user space, and uses transport-layer protocols (UDP) for communication rather than directly forwarding packets like network-layer protocols.

🧠 The Correct Way to Think About Layers

❌ WRONG thinking:

“DHCP deals with IP, so it must be network layer”

✅ RIGHT thinking:

Layers are defined by *what a protocol does*, not *what it configures*

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📊 What the Network Layer Does (Layer 3)

Network layer protocols:

- Move packets from source to destination
- Perform routing and forwarding
- Example:
 - IP
 - ICMP
 - Routing protocols (conceptually L3)

📌 Network layer does packet delivery, not configuration.

📦 What DHCP Actually Does

DHCP:

- Assigns IP addresses
- Assigns subnet mask
- Assigns gateway & DNS
- Manages leases

👉 These are services provided to hosts, not packet forwarding.

📌 DHCP does not route packets

📌 DHCP does not forward traffic

📌 DHCP does not examine routing tables

Layer-wise Breakdown of DHCP

Layer	Role in DHCP
Application	Defines DHCP messages (Discover, Offer, Request, ACK)
Transport	Uses UDP (ports 67 & 68)
Network	Uses IP (broadcast 255.255.255.255)
Link	Uses MAC broadcast

👉 DHCP uses lower layers but does not belong to them.

⚙️ Key Reason #1: DHCP Runs in User Space

- DHCP server = application software
- DHCP client = application process

Examples:

- `dhcpcd` on Linux
- Windows DHCP Client service

📌 Network-layer protocols are implemented inside the OS kernel, DHCP is not.

🔄 Key Reason #2: DHCP Uses UDP (Not Part of It)

A protocol that:

- Uses TCP or UDP
- Runs above transport layer

➡ Must be Application Layer

Same logic applies to:

- HTTP
- FTP
- DNS
- SMTP

📌 DHCP uses UDP, therefore it is above transport layer.

🔄 Key Reason #3: Client-Server Model

DHCP follows:

- Client-server architecture
- Request-response messaging
- Application-level logic (leases, timers, policies)

📌 Network layer is stateless, DHCP is stateful.

🔧 Key Reason #4: Configuration ≠ Communication

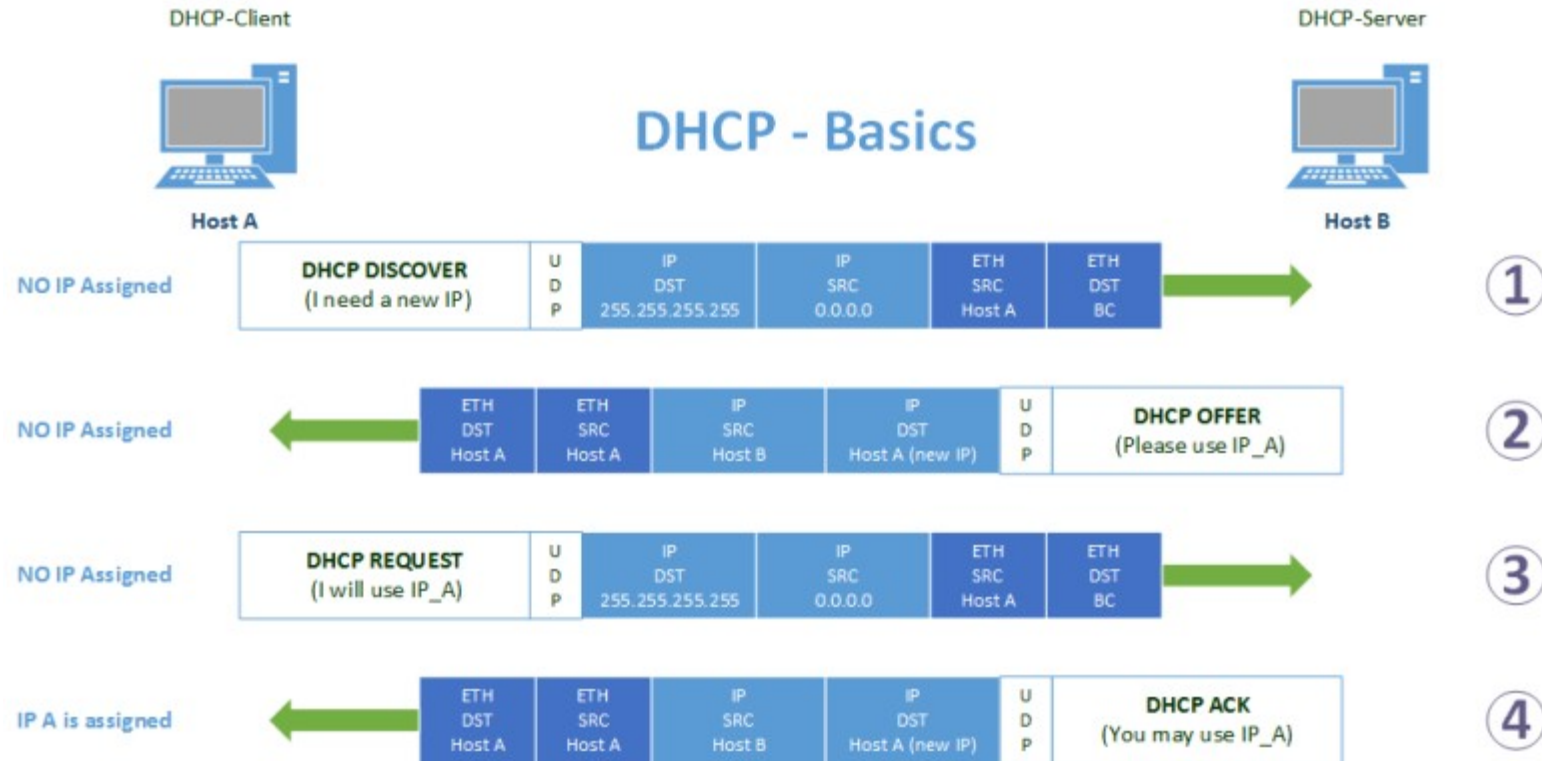
DHCP:

- Helps a host prepare for network communication
- Does not participate in actual data transfer

Think of it as:

“Setup assistant” for networking

DHCP: Layering



<https://crnetpackets.com>

DHCP: Encapsulation

- A DHCP message is inside a UDP segment, which is inside an IP packet.
 - *Encapsulation layers:* The process follows the standard network model, with DHCP at the application layer.
 - DHCP messages are sent within a User Datagram Protocol (UDP) segment.
 - The UDP segment is then placed inside an Internet Protocol (IP) packet.
 - *Why this order:*
 - UDP: DHCP relies on UDP for speed rather than guaranteed delivery.
 - IP: The IP packet provides the source and destination IP addresses to route the message across networks.

DHCP: Encapsulation

